

Spring Challenge 2026 - Troll Farm

Welcome to the Bronze league!

In Bronze you can now **CHOP** trees for wood and **MINE** iron. Trees grow faster near water. Games last **300** turns.

There will be no additional rules in higher leagues.

The Goal

You control a pack of trolls. Make them collect the most resources.

Rules

Each player starts with a shack and a troll. Control your troll to collect fruits, mine iron and chop trees and then bring the resources back to your shack. The resources can be used to train more trolls or plant trees. The game is played on a grid, cells have different terrains and trees on them.

Troll attributes

Each troll has **4** attributes: **movementSpeed**, **carryCapacity**, **harvestPower** and **chopPower**. They can't be changed.

Trees

A tree has a **type** (**PLUM**, **LEMON**, **APPLE**, **BANANA**), a **size**, a **health**, an amount of **fruits** and a **cooldown**. When cooldown reaches **0**, the tree either grows in size or, if at maximum size (**4**), produces fruits. Trees can hold up to **3** fruits.

Trees grow faster when next to water.

Trees differ in cooldown and health as follows:

	PLUM	LEMON	APPLE	BANANA
Cooldown	8	8	9	6
Cooldown near water	3	3	2	4
Health (size 1)	6	6	11	3
Health (size 2)	8	8	14	4
Health (size 3)	10	10	17	5
Health (size 4)	12	12	20	6

When a damaged tree grows, it gains the health difference between sizes.

Moving units

Each troll has a **movementSpeed** determining how many cells it can **MOVE** horizontally or vertically per turn. Each cell can hold at most **1** troll per team. Only **GRASS** cells are walkable. The shack is also the spawn point, but trolls can't walk back on that cell after leaving it. If the target is out of range or not walkable, the troll moves to the nearest reachable cell towards it.

Harvesting

Trees can be harvested, given they have fruits on them. When a troll shares a cell with a tree, **HARVEST** will take as many fruits as possible by free **carryCapacity** and **harvestPower**. When two trolls harvest the same tree in parallel, they keep taking one fruit at a time, as long as there are fruits left and they can still harvest. The last fruit can get duplicated, so that both trolls can have it.

Planting trees

A troll can **PLANT** a tree on its current cell if the troll carries at least **1** fruit of that type. If two trolls try to plant a tree of the same type on the same cell, both will lose a seed and the tree gets planted. If they try to plant different types, nothing will happen.

Chopping trees

A troll can **CHOP** a tree on the same cell, reducing its health by **chopPower**. When health reaches **0**, the tree disappears and the troll collects wood equal to its size, capped by free **carryCapacity**. Any wood that the troll can't carry will just vanish. Similar to harvesting, both trolls get one wood at a time until they can't carry any more or the wood is completely split up. The last wood can get duplicated.

Mining

A troll adjacent to an **IRON** cell can **MINE**, gaining up to **chopPower** iron limited by free **carryCapacity**. Iron resources are infinite.

Dropping items

When a troll carrying resources is next to its shack (horizontally or vertically), **DROP** transfers all carried items to the shack.

Picking items

A troll can **PICK** one item from its shack. Only **1** item can be picked at a time.

Training trolls

Use **TRAIN** to spawn a new troll at your shack and set its four attributes. Each attribute requires a different resource:

- **PLUM** for **movementSpeed**
- **LEMON** for **carryCapacity**
- **APPLE** for **harvestPower**
- **IRON** for **chopPower**

The cost for each attribute is equal to the number of existing trolls in the own team plus the attribute squared.

Example: you already have **2** trolls and want to **TRAIN** the following, giving you these costs:

- **2 movementSpeed**: $2 + 2 \times 2 = 6$ **PLUMs**
- **3 carryCapacity**: $2 + 3 \times 3 = 11$ **LEMONs**
- **1 harvestPower**: $2 + 1 \times 1 = 3$ **APPLEs**
- **0 chopPower**: $2 + 0 \times 0 = 2$ **IRONs**

Victory Conditions

- You score more points than your opponent. Each fruit in your shack scores **1** point; **WOOD** scores **4** points. **IRON** has no score value

Loss Conditions

- You have fewer points than your opponent
- You fail to respond in time or output an unrecognized command

Expert Rules

Referee source code: [source code](#)

Don't hesitate to change the viewer's options to help debug your code.

The game turn order is:

1. Move trolls
2. Harvest
3. Plant
4. Chop
5. Pick
6. Train
7. Drop
8. Mine
9. Grow trees

Actions of the same type occur simultaneously. If both teams harvest or chop the same tree at once, resources are shared equally.

Then both get one item at a time until they can't carry any more or the resource is exhausted. The last item can get duplicated.

Each troll can only perform one action per turn.

Game Input

Initial input

- **First line**: **width height**, the grid size
- **Next height lines**: each line has **width** characters: **.** for **GRASS**, **~** for **WATER**, **#** for **ROCK**, **+** for **IRON**, **0** for your own **SHACK**, **1** for your opponent's **SHACK**

Input for each game turn

- **First two lines:** your inventory followed by your opponent's inventory: `plums, lemons, apples, bananas, iron, wood`
- **Next line:** `treeCount`, the number of trees
- **Next `treeCount` lines:** `type x y size health fruits cooldown`, where `type` can be **PLUM, LEMON, APPLE, BANANA**
- **Next line:** `trollsCount`, the number of trolls
- **Next `trollsCount` lines:** `id player x y movementSpeed carryCapacity harvestPower chopPower carryPlum carryLemon carryApple carryBanana carryIron carryWood`, where `player` is **0** if you own the troll, **1** otherwise

Output for one game turn

- Each turn you can print any number of commands, separated by ;
 - `MOVE id x y` Move troll `id` to cell (`x`, `y`)
 - `HARVEST id` Make troll `id` harvest on its current cell
 - `PLANT id type` Make troll `id` plant a `type` on its current cell: **PLUM, LEMON, APPLE or BANANA**
 - `CHOP id` Make troll `id` chop on its current cell
 - `PICK id type` Make troll `id` pick one `type` from the shack: **PLUM, LEMON, APPLE or BANANA**
 - `DROP id` Make troll `id` drop all carried items at the shack
 - `TRAIN moveSpeed carryCapacity harvestPower chopPower` Train a new troll with the given attributes
 - `MINE id` Make troll `id` mine a nearby **IRON**
 - `WAIT` to do nothing
 - `MSG text` to display a message in the replay

Constraints

- Response time for the first turn ≤ 1000 ms
- Response time per turn ≤ 50 ms

Note: you will only lose the game when you exceed the time limit 3 times by at most 50 ms or once by more than that. This is to counter random delays when servers are busy, it is not advised to intentionally exceed the limits.

- The game ends after **300** turns, when either player can enforce a win by doing nothing or when there are no trees on the map for **10** consecutive turns
- $8 \leq \text{height} \leq 11$
- $\text{width} = 2 * \text{height}$